Python Programming Notes

1. What is Python?

Python is a high-level, interpreted programming language. It is widely used in web development, data science, artificial intelligence, machine learning, and automation.

Python has a simple syntax that makes it easy for beginners to learn.

2. Variables in Python

A variable is a name used to store a value.

Example:

```
name = "Ayush"

age = 20

branch = "CSE"


print(name)

print(age)

print(branch)
```

In this example, name stores a string, age stores an integer, and branch stores a string.

3. Data Types in Python

Common Python data types include:

- int: Stores integer values, such as 10 or 25.
- float: Stores decimal values, such as 3.14.
- str: Stores text, such as "Hello".
- bool: Stores True or False.
- list: Stores an ordered collection of items.
- tuple: Stores an ordered collection that cannot be modified.
- dict: Stores data in key-value pairs.
- set: Stores unique values.

4. Functions in Python

A function is a reusable block of code that performs a particular task.

Example:

```
def add(a, b):
    return a + b
```

```
result = add(10, 20)

print(result)
```

Output:

```
30
```

5. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

Applications include chatbots, recommendation systems, image recognition, and speech recognition.

6. What is RAG?

RAG stands for Retrieval-Augmented Generation.

RAG combines information retrieval with text generation. It allows an AI application to retrieve relevant information from external documents and use that information to generate an answer.

The main stages of a RAG system are:

1. Load the documents.
2. Split the documents into smaller text chunks.
3. Convert the chunks into embeddings.
4. Store the embeddings in a vector database.
5. Retrieve relevant chunks when a user asks a question.
6. Use a language model to generate an answer based on the retrieved information.

7. Benefits of RAG

- It allows an AI application to use custom documents.
- It can provide answers grounded in retrieved information.
- It allows document collections to be updated without retraining the language model.
- It can provide references to the source documents.

8. What is a Vector Database?

A vector database stores and searches numerical vector representations of information.

In a RAG application, document chunks are converted into embeddings. The system searches these embeddings to find information relevant to a user's question.

FAISS is a library that supports efficient similarity search over vectors.